

Question number	Answer	Additional guidance	Mark
4(a)	A description that includes two of the following points: - as intensity increases heart rate increases (1) - smaller effect at {low intensity / high intensity} (1)	Positive correlation ALLOW Stated comparative effect eg. Largest increase between 6-8 au	Graduate (2)

Question number	Answer	Additional guidance	Mark
4(b)	An answer showing the following steps: - calculation of heart rates (1) - calculation of change in heart rate (1) - correct number of decimal places and units (1)	ECF for mp1 correct calculation in cm³ Example of calculation: $4.43 \div 0.0744 = 59.543$ $4.21 \div 0.0584 = 72.089$ $72.1 - 59.5 = 12.546$ ECF mp2 subtraction (from mp1) and correct number of d.p Answer= 12.55 (b)pm Correct answer with units – 3 marks	Expert (3)

Question number	Answer	Mark
4(c)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive, and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. Indicative content The first table shows • as exercise intensity increases heart rate increases The second table shows • regular exercise increases stroke volume • regular exercise has little effect on resting cardiac output • meaning heart rate is slower Own knowledge • during exercise muscles use energy / ATP • most of this energy is produced via aerobic respiration • reference to increased carbon dioxide from increased cell respiration • effect of increased carbon dioxide in blood eg. – pH sensors in carotid arch • heart (and circulation) provides bulk transport of oxygen and {nutrients / glucose} • increased demand requires increased blood circulation • stroke volume and rate of beating determine volume of blood moved • regular exercise increases stroke volume so greater capacity to respond by increasing heart rate	Expert (6)

Level	Marks	Descriptor
	0	No awardable content.
1	1-2	An explanation may be attempted but with limited interpretation or analysis of the scientific information and with focus on mainly just one piece of scientific information. The explanation will contain basic information, with some attempt made to link knowledge and understanding to the given context.
2	3-4	An explanation will be given, with occasional evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows some linkages and lines of scientific reasoning with some structure.
3	5-6	An explanation is made that is supported throughout by sustained application of relevant evidence of analysis interpretation and/or evaluation of both pieces of scientific information. The explanation shows a well-developed and sustained line of scientific reasoning, which is clear and logically structured.